

ASSESSMENT OF GROUNDWATER QUALITY FOR DRINKING PURPOSES IN THE INDUSTRIAL CORRIDOR OF THE SONIPAT DISTRICT ALONG NH-44, HARYANA, INDIA

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ABSTRACT

The rapid expansion of industrial and agricultural activities has increased pressure on natural water resources, making continuous monitoring of groundwater quality critically important. The present study was conducted to assess the suitability of groundwater for drinking purposes in the HSIIDC (Haryana State Industrial and Infrastructure Development Corporation) industrial regions of Sonipat district. Groundwater samples were collected from ten industrial locations within the HSIIDC industrial estates of Sonipat City, Barhi, Kundli, Murthal and Rai along NH-44 (National Highway-44) corridor in Sonipat district. Samples were collected in triplicate at each site and the mean values were used for groundwater quality assessment. A total of sixteen physicochemical and biological water quality parameters were analysed, including pH, total dissolved solids (TDS), total hardness (as CaCO₃), calcium (Ca²⁺), magnesium (Mg²⁺), carbonate (CO₃²⁻), bicarbonate (HCO₃⁻), sodium (Na⁺), potassium (K⁺), nitrate (NO₃⁻), sulphate (SO₄²⁻), chloride (Cl⁻), fluoride (F⁻), dissolved oxygen (DO), biochemical oxygen demand (BOD) and chemical oxygen demand (COD). All analyses were conducted using standard procedures given by the American Public Health Association (APHA) and compared with the permissible limits established by the World Health Organization (WHO) and the Bureau of Indian Standards (BIS) IS 10500:2012.

The results indicate that the concentrations of calcium, carbonates, bicarbonates, sodium, nitrate, chloride and fluoride were within the permissible limits recommended by the BIS and the WHO guidelines at most sampling locations. However, localized exceedances and elevated concentrations of certain parameters at specific sites indicate potential risks associated with industrial influence and groundwater contamination. Elevated values of pH, TDS and total hardness were observed at some locations. According to the Water Quality Index (WQI), approximately 80 % of the samples were within the “good” category, with the exception of sample site S9 (Barhi) which showed poorer water quality and required purification before consumption. Spatial variations in groundwater quality were observed across the study area.

Keywords: water quality, physicochemical parameter, Sonipat, groundwater, TDS

INTRODUCTION

Water is a fundamental necessity for humans, plants and animals. About 30 % of the world's freshwater supply is provided by groundwater. When accounting for oceans and ice, groundwater makes up approximately 0.76 % of the world's total water. In various developing countries like India, Pakistan, China and Nepal, a large number of people depend on the groundwater for their basic needs, especially for drinking purposes [1, 2]. The quality of water is deteriorating day by day. The main factors responsible for the deterioration of groundwater are the rapid growth of the population, uncontrolled industrialisation, excessive use of fertilisers, urbanisation and ineffective groundwater management practices [3 - 6].

Waste generated from human activities, particularly due to heavy industrialization, can seep into groundwater and surface water, thereby making both potentially hazardous for human consumption. Consumption of contaminated groundwater has a negative impact on health, leading to skeletal fluorosis and dental issues like discolouration of the tooth enamel [7]. The treatment of industrial waste before its release into water bodies or disposal, the implementation of a proper drainage system and effective waste management practices can reduce the risk of groundwater contamination.

Rapid industrialisation is a cornerstone of economic growth, but it often comes at the cost of environmental degradation, particularly impacting water resources. Industrial areas characterised by a large number of diverse manufacturing and processing activities are major sources of pollutants that can compromise water quality. These pollutants, including heavy metals, organic compounds and other chemicals, can leach into surface and groundwater sources, posing significant risks to human health and ecological integrity. Understanding the nature and extent of industrial impacts on water resources is, therefore, crucial for sustainable development and effective environmental management [8].

Due to the excessive withdrawal of groundwater, various countries are suffering from water scarcity. In the present scenario, effective management of both the quality and quantity of groundwater resources has become a major environmental challenge. Thus, it is essential to develop a scientific understanding of water quality for effective management.

The Kundli, Rai, Bahalgarh, Murthal and Barhi industrial areas, located in Haryana (India), are significant hubs of industrial activity. The region hosts a variety of industries, including manufacturing, textiles, food processing and others, which make a significant contribution to the economic landscape of the state. However, this concentration of industrial activity raises concerns about the potential for water pollution, given the wastewater generated by these industries and the hydrogeology of these areas. The groundwater in this area serves as a primary source for drinking and other domestic purposes, highlighting the need for a comprehensive investigation into the current water quality status.

This study focuses on analysing the water quality in selected industrial locations. The aim of this study is to assess the levels of various physicochemical and biological parameters in groundwater samples to determine the extent of pollution and identify its potential sources. The findings of this research will provide valuable information on the environmental impact of industrial operations in this region and will contribute to the development of effective strategies for pollution control, water resource management and public health protection. This study aims to assess the current status of water quality and the levels of the contaminants of greatest concern.

MATERIALS AND METHODS

Study area

Sonipat is a district of Haryana, India, located within the National Capital Region (NCR). It

covers an area of 2260 km² with a population density of 640 people per km². It lies 20 km north of New Delhi, the capital of India. Sonipat is bordered by Jind to the north-west, Panipat to the north, Rohtak to the west, and Uttar Pradesh to the east, with the Yamuna River forming a natural boundary. The major national highways crossing the district are NH-1, NH-44, NH-71, NH-334B, and the Eastern and Western Peripheral Expressways.

Groundwater resources in Sonipat vary from area to area. The depth of the water table is the shallowest along the Yamuna River, where it is approximately 3 m, and increases to 9 - 12 m in southeastern and western parts of Sonipat. In this region, groundwater exhibits salinity and has a very high fluoride content. The major industrial estates in Sonipat are Sonipat City, Barhi, Kundli, Murthal and Rai (Figure 1).

Data collection

Groundwater samples were collected from ten selected industrial locations representing the major industrial clusters of Kundli, Rai, Bahalgarh, Murthal and Barhi along the NH-44 corridor in the Sonipat district. Site selection was based on industrial intensity and

spatial distribution to ensure adequate coverage of the study area. Although the number of sampling locations is limited, these sites provide a reasonable representation of groundwater quality in the industrial corridor.

Water samples were carefully collected in pre-cleaned, high-density polyethylene (HDPE) bottles of 2-L capacity, following the APHA guidelines. The bottles were rinsed three times with the sample before final sample collection to avoid contamination. After collection, the samples were immediately stored in insulated containers and transported to the laboratory under cooled conditions at a temperature of approximately 4 °C. Physicochemical parameters such as pH and DO were measured within 6 hours of sampling, while other chemical parameters were analysed within 24 - 48 hours, according to the holding time recommended by the American Public Health Association (APHA, 2017). At each sampling location, water samples were collected in triplicate to ensure the accuracy and reliability of the analytical measurements. The instruments were calibrated using standard solutions. The three replicate readings for each parameter were averaged, and the resulting mean values were used for data analysis.

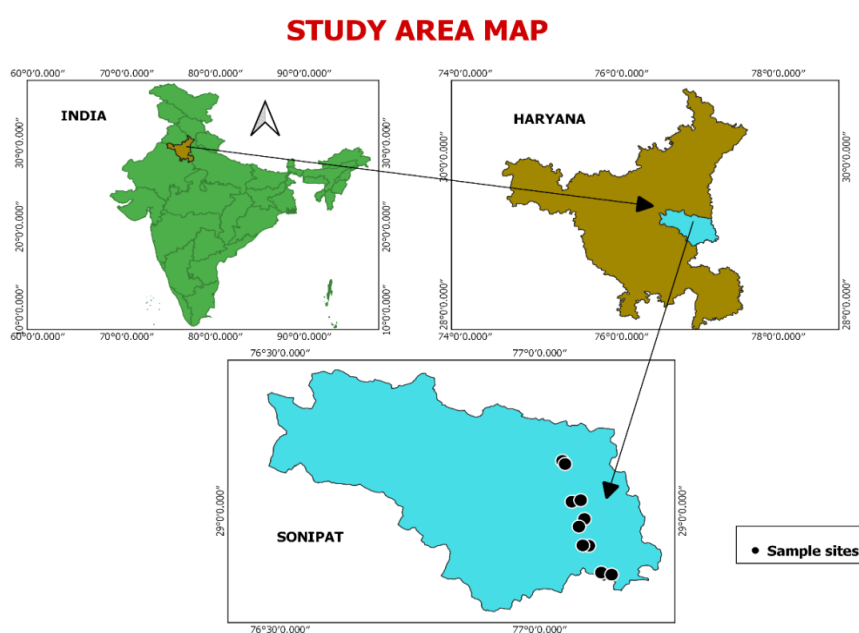


Figure 1. Location map of the study area, Sonipat (Haryana), showing the sampling sites

In the present study, physicochemical and biological analyses of groundwater from various locations within the industrial areas of Kundli, Rai, Bahalgarh, Murthal, Bhigan and Barhi were carried out. The industrial sites under study, along with their geographical coordinates, are shown in Table 1.

Table 1. The sampling sites and their geographical coordinates

Sample No.	Industries	Latitude (N)	Longitude (E)
S1	Nandi Polytube sec. 53 HSIIDC Kundli	28°52'3.65"	77°01'5.03"
S2	JSG Innotech Pvt. Ltd. phase 5 Kundli	28°52'10.01"	77°08'19.68"
S3	Plot 1915 Fiem industries Ltd. Phase -5 Rai	28°55'42.80"	77°05'41.29"
S4	Kayem food industries Pvt. Ltd. sec 38 Rai	28°55'46.19"	77°04'54.55"
S5	4/25 Suman textiles sec 29 Bahalgarh	28°58'8.63"	77°05'8.62"
S6	Jasch industries Ltd. Sec. 28 Bahalgarh	28°58'20.09"	77°04'28.24"
S7	Brake Parts India Pvt. Ltd. Vill. Rewli Murthal	29°01'7.64"	77°03'42.13"
S8	Coral drugs Pvt. Ltd. HSIIDC Murthal	29°01'17.61"	77°04'3.19"
S9	412, A.S. Enterprises, Phase I, HSIIDC Barhi	29°06'11.67"	77°02'37.17"
S10	384, Samarth Industries HSIIDC Barhi	29°05'55.21"	77°02'48.84"

The determination of pH (measured as the power of hydrogen, $-\log[\text{H}_3\text{O}^+]$), total dissolved solids (TDS), total hardness (TH) as calcium carbonate (CaCO_3), calcium (Ca^{2+}), magnesium (Mg^{2+}), carbonate (CO_3^{2-}), bicarbonate (HCO_3^-), sodium (Na^+), potassium (K^+), nitrate (NO_3^-), sulphate (SO_4^{2-}), chloride (Cl^-) and fluoride (F^-) was carried out in the research laboratory at Baba Mastnath University (BMU), Rohtak. The analysis of dissolved oxygen (DO), biochemical oxygen demand (BOD) and chemical oxygen demand (COD) was carried out in the Envirochem Testing Laboratory and Research Centre, Panipat. Groundwater quality parameters were

analysed using standard methods shown in Table 2, according to the APHA protocol [9].

Table 2. Analytical methods used for assessing the water quality parameters

Sr. No.	Parameter	Analytical method	Analytical technique
1.	pH	Electrometric	pH meter
2.	TDS	Filtration	Filtration and drying
3.	TH	EDTA Titrimetric method	Complexometric titration
4.	Ca^{2+}	EDTA Titrimetric method	Complexometric titration
5.	Mg^{2+}	Calculated from TH and Ca	Calculation based
6.	CO_3^{2-} and HCO_3^-	Acid-Base titration	Titration with H_2SO_4
7.	Na^+ and K^+	Flame photometric method	Flame photometry
8.	NO_3^-	UV spectrophotometric method	Spectrophotometry
9.	SO_4^{2-}	Turbidimetric method	Turbidity meter
10.	Cl^-	Argentometric method	Silver nitrate titration
11.	F^-	UV spectrophotometric method	Spectrophotometry
12.	DO	Modified Winkler's method	Winkler's method
13.	BOD	Modified Winkler's method	Winkler's method
14.	COD	Dichromate reflux method	Closed reflux digestion followed by spectrophotometric determination

Water quality standards

The World Health Organization (WHO) released the fourth edition of its Guidelines for Drinking Water Quality (GDWQ) in 2017 to create a detailed framework for evaluating and managing the safety of drinking water globally [10]. In India, the Bureau of Indian Standards (BIS) introduced the IS 10500:2012 specifications, which serve as the primary

regulatory framework for drinking water quality [11]. These guidelines set permissible limits for a range of physical, chemical and microbiological parameters to ensure that water is safe and suitable for consumption. The BIS 2012 standards were developed to address growing concerns over water contamination and its impact on human health. They provide two sets of limits: acceptable limits, which indicate safe consumption levels, and permissible limits, which specify the maximum allowable concentrations in the absence of an alternative water source. The standards cover a range of contaminants, including microorganisms, heavy metals, pesticides and dissolved solids, to prevent health hazards such as waterborne diseases, toxicity and long-term illnesses. Table 3 shows the standard values of the 13 physicochemical parameters according to the BIS (2012) and WHO (2017) standards. The determined water quality parameters were compared with the standard values established by the WHO and BIS.

Water quality index (WQI) and its mathematical formulation

WQI is a quantitative measure that provides a single number to evaluate the overall water quality, taking into account various physical,

chemical and biological parameters. It condenses complex water quality data into a single numerical value, which facilitates the communication of water body health [12 - 14]. WQI is widely used for monitoring surface water, groundwater and drinking water sources, helping policymakers and researchers to identify pollution trends and necessary interventions. WQI does not include all parameters, rather, it is based on those that serve as key indicators of water pollution levels, suitability for drinking, and other uses.

WQI was computed using the weighted arithmetic water quality index (WAWQI) method, which has been widely adopted by numerous researchers [15 - 17]. This method was selected due to its reliability and wide application in groundwater quality studies. It involves assigning unit weights based on the permissible limits of the parameters according to the BIS and WHO standards, ensuring that the parameters with greater health significance exert a stronger influence on the final WQI value. It is calculated using the equation given by Brown et al. in 1972 [18]:

$$\text{WAWQI} = \Sigma(Q_n \cdot W_n) / \Sigma W_n \quad (1)$$

where Q_n is the quality rating of the n^{th} parameter and W_n is weight of the n^{th} parameter.

Table 3. Drinking water quality guidelines (BIS and WHO)

Sr. No.	Physicochemical parameters	BIS, 2012 standards		Standard limits according to WHO (2017) guidelines
		Desirable/Acceptable limit	Permissible limit	
1.	pH	6.5 - 8.5	No relaxation	6.5 – 8.5
2.	TDS	500	2000	≤ 1000
3.	TH as CaCO ₃	200	600	500
4.	Ca ²⁺	75	200	75
5.	Mg ²⁺	30	100	50
6.	CO ₃ ²⁻	75	200	-
7.	HCO ₃ ⁻	200	600	-
8.	Na ⁺	-	-	< 200
9.	K ⁺	-	-	10 - 12
10.	NO ₃ ⁻	45	No relaxation	50
11.	SO ₄ ²⁻	200	400	-
12.	Cl ⁻	250	1000	200 - 300
13.	F ⁻	1.0	1.5	1.5

all concentrations are expressed in mg/L, except pH (dimensionless)

The Q_n is calculated using the following equation:

$$Q_n = [(C_n - C_{ideal}) / (S_n - C_{ideal})] \cdot 100 \quad (2)$$

where C_n is the experimentally observed value of the n^{th} parameter, C_{ideal} is the ideal value of the n^{th} parameter, which is zero for all parameters with the exception of pH (7.0) and DO (14.6 mg/L), and S_n is the standard permissible value for the n^{th} parameter.

The W_n is calculated using the following equation:

$$W_n = k/S_n \quad (3)$$

where k is a constant of proportionality determined using the following equation:

$$k = 1 / \sum(1/S_n) \quad (4)$$

In the present study, the sum of the unit weights ($\sum W_n$) was normalised to unity. Therefore, the calculated value of $\sum(Q_n \cdot W_n)$ directly represents the final water quality index (WQI) for each sampling location.

The standard classification of water quality based on WQI values is presented in Table 4.

Table 4. Water quality classification based on WQI values [19]

WQI range	Water quality status	Grading
0 - 25	Excellent (Safe for drinking)	A
26 - 50	Good (Acceptable)	B
51 - 75	Poor (Needs treatment)	C
76 - 100	Very poor (Unsuitable)	D
> 100	Unfit for drinking	E

Correlation of groundwater quality parameters

The correlation coefficient (r) is a statistical metric that assesses the strength and direction of the association between two variables [20, 21]. It has a numerical value ranging from -1 to +1, indicating how strongly two variables

are related. A correlation coefficient value approaching -1 or +1 indicates a strong correlation between the two parameters [22, 23]. A positive r value suggests that an increase in one parameter leads to an increase in the related parameter, while a negative r value indicates that an increase in one parameter leads to a decrease in the related parameter. A value of $r > 0.7$ indicates a strong correlation between the two parameters. An r value between 0.5 and 0.7 indicates a moderate correlation, while an r value approaching zero represents no significant correlation between the parameters.

Pearson's correlation coefficients (r) were calculated using the mean values of triplicate measurements for 10 sampling sites ($n = 10$). The statistical significance of correlations was evaluated using a two-tailed t -test. Correlations were considered significant at $p < 0.05$ and highly significant at $p < 0.01$.

RESULTS AND DISCUSSION

Statistical data on the physicochemical parameters of the ten groundwater samples collected from the industrial locations of the Sonipat district located along NH-44 are presented in Table 5 and illustrated in Figure 2.

Physicochemical parameters of water

pH

pH acts as a critical parameter for ensuring water quality. It provides a measurement of the acidity or alkalinity of water. According to BIS standards (Table 3), water is fit for drinking when the pH ranges between 6.5 and 8.5, and unfit when the pH is below 6.5 or above 8.5. The pH value of the water samples from different locations ranges from 7.61 to 8.66, indicating a generally neutral to slightly alkaline nature. The pH of the water from nine sample locations is within desirable limits, while the water from site S9 (Barhi) is non-potable due to a pH of 8.66, indicating the

presence of carbonate and bicarbonate ions, industrial discharge, etc. Acidic water can cause corrosion, while higher values can lead

to scaling in pipelines and water distribution systems due to CaCO_3 precipitation.

Table 5. Physicochemical analysis data for groundwater samples from the Sonipat industrial areas

Location	pH	TDS	TH	Ca^{2+}	Mg^{2+}	CO_3^{2-}	HCO_3^-	Na^+	K^+	NO_3^-	SO_4^{2-}	Cl^-	F^-	DO
Nandi Polytube sec. 53 HSIIDC Kundli	7.61	1724	612	101	89.6	0.6	538	328	21.1	8.8	304	362	0.29	4.8
JSG Innotech Pvt. Ltd. phase 5 Kundli	7.8	1185	890	160	144.1	0.8	420	289	19.4	9.7	431	213	0.25	4.8
Plot 1915 Fiem industries Ltd. Phase -5 Rai	7.74	4765	1165	188	164	0.5	394	864	30	9.62	791	856	0.34	4.7
Kayem food industries Pvt. Ltd. sec 38 Rai	7.67	2923	850	142	123.2	0.7	282	624	13	8.20	530	624	0.18	6.4
4/25 Suman textiles sec 29 Bahalgarh	8.27	3132	490	72	70.7	11.4	290	774	273	9.1	456	660	0.26	4.3
Jasch industries Ltd. Sec. 28 Bahalgarh	7.96	952	485	100	58.5	0.4	336	58.2	18.5	7.64	93	108	0.24	3.9
Brake Parts India Pvt. Ltd. Rewli Murthal	7.78	852	455	71.7	68.8	0.5	433	68	4	2.94	86.2	158	0.64	7.9
Coral drugs Pvt. Ltd. HSIIDC Murthal	7.83	926	510	85.3	73.5	0.8	442	64	13	3.21	96.6	166	0.92	8.1
412, A.S. Enterprises, Phase I, HSIIDC Barhi	8.66	788	256	52.8	30.9	31.5	353	87	7	3.41	148	157	0.82	8.2
384, Samarth Industries HSIIDC Barhi	8.24	448	204	52	18.4	5.7	336	75.4	2.7	4.91	16.3	27	0.22	5.2

where is: pH - potential of hydrogen (dimensionless), TDS - total dissolved solids (mg/L), TH - total hardness (as CaCO_3) (mg/L), Ca^{2+} - calcium (mg/L), Mg^{2+} - magnesium (mg/L), CO_3^{2-} - carbonate (mg/L), HCO_3^- - bicarbonate (mg/L), Na^+ - sodium (mg/L), K^+ - potassium (mg/L), NO_3^- - nitrate (mg/L), SO_4^{2-} - sulphate (mg/L), Cl^- - chloride (mg/L), F^- - fluoride (mg/L) and DO - dissolved oxygen (mg/L).



Figure 2. Variation in physicochemical water parameters at ten locations within the industrial areas of Sonipat

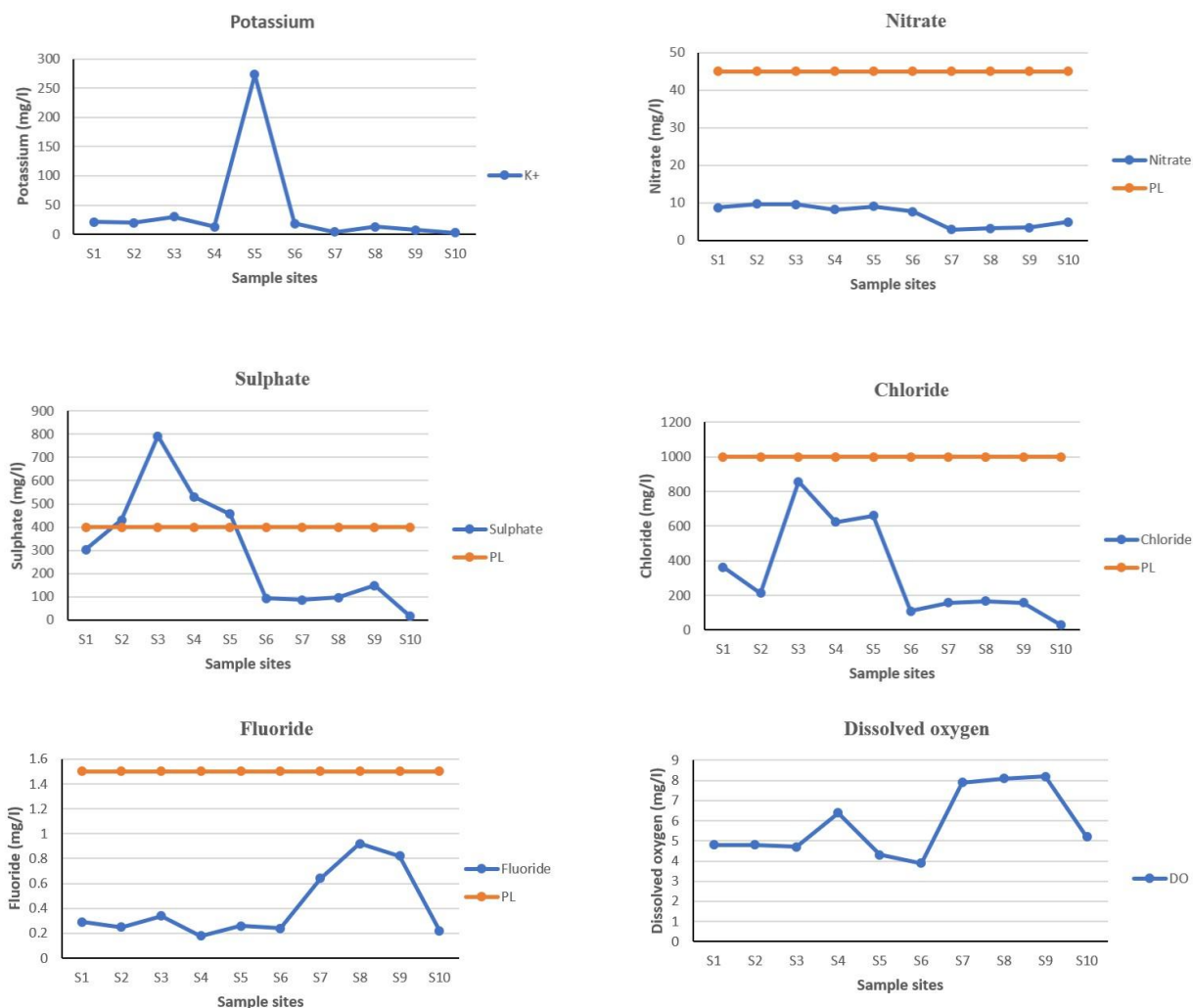


Figure 2 (continued). Variation in physicochemical water parameters at ten locations within the industrial areas of Sonipat

TDS (total dissolved solids)

TDS is a crucial parameter that provides information on the concentration of substances dissolved in water, such as minerals from rocks and soil, organic matter and salts. Underground water having a TDS level of up to 500 mg/L is desirable for drinking purposes, while a range of 500 - 2000 mg/L (Table 3) lies within the permissible limits. The TDS values in the collected samples varied from 448 mg/L to 4765 mg/L. TDS values at seven locations are within the permissible limits. The samples from locations S3 (4765 mg/L), S4 (2923 mg/L) and S5 (3120 mg/L) have very high TDS values, which make them unfit for drinking due to the excessive presence of dissolved salts or contaminants. Such high TDS levels may significantly reduce

palatability and can cause gastrointestinal discomfort, in addition to promoting scaling in household plumbing and industrial systems.

TH (total hardness)

TH is the total concentration of Ca^{2+} and Mg^{2+} ions in water, expressed in mg/L as CaCO_3 . Total hardness in the water samples from the study area varied from 204 mg/L to 1165 mg/L, exceeding the permissible limit of 600 mg/L. Site S3 (Rai industrial area, phase-5), contains the highest TH level (1165 mg/L), indicating excessive mineral content, likely from geological sources, groundwater leaching, or limestone deposits. It requires extensive water softening and treatment before industrial use. Hard water can cause scaling in

pipelines and industrial equipment, which increases maintenance costs and reduces the efficiency of machinery. The hardness of the water samples is within the permissible limits at 6 locations, while the water is non-potable at 4 locations situated in HSIIDC Kundli and HSIIDC Rai. The elevated TDS and hardness observed at sites closer to Delhi may be attributed to intensive urbanization, industrial activities, and the dissolution of carbonate minerals present in the alluvial sediments of the region.

Textile, paper, food processing and pharmaceutical industries require soft water, making the water from these locations unsuitable without treatment. Samples S9 and S10 have the most suitable water quality for drinking and industrial use.

Calcium and magnesium (Ca^{2+} and Mg^{2+})

Accounting for 3.63 % of the Earth's crust, calcium is the fifth most abundant element. It is a crucial element required for cellular physiology and bone health [24]. Approximately 99 % of the body's calcium is stored in the teeth and bones. According to the BIS standards, the desirable limit for Ca^{2+} is 75 mg/L and the maximum permissible limit is 200 mg/L, while the desirable limit for Mg^{2+} is 30 mg/L and the maximum permissible limit is 100 mg/L. Magnesium is an essential element that plays a vital role in the proper functioning of living organisms, particularly for the cardiac muscle and kidneys. The calcium ions in samples S5, S7, S9 and S10 are within the desirable limit, while the concentrations in the remaining samples lie within the permissible limit. The concentration of magnesium ions exceeds the desirable limit (30 mg/L) in all samples except S10 (Barhi). The concentrations of calcium ions range from 52 to 188 mg/L, and those of magnesium ions from 18.4 to 164 mg/L. Elevated magnesium concentrations at certain locations may pose potential health risks with long-term consumption and indicate a localized deterioration in groundwater quality.

Carbonates and bicarbonates (CO_3^{2-} and HCO_3^-)

Carbonate (CO_3^{2-}) and bicarbonate (HCO_3^-) ions are components of water chemistry that influence alkalinity, buffering capacity and pH regulation. Their presence in natural water sources is governed by interactions with carbonate minerals, the dissolution of atmospheric CO_2 , and anthropogenic factors. The carbonate ion concentration ranges from 0.4 to 31.5 mg/L, while the bicarbonate concentration exhibits a broad range, from 282 to 538 mg/L. The highest bicarbonate concentration is observed at S1 (538 mg/L). Samples S2, S3, S7 and S8 also have high bicarbonate levels (394 - 442 mg/L), indicating a bicarbonate-dominated buffering system. Although carbonate and bicarbonate ions contribute to the natural buffering capacity of groundwater, elevated bicarbonate concentrations may increase alkalinity and affect the palatability and suitability of water for long-term domestic use and consumption.

Sodium and potassium (Na^+ and K^+)

Sodium is primarily found in extracellular fluids and is essential for electrolyte balance and muscle contraction, while potassium is the major intracellular cation and is vital for regulating cardiac function. The proper balance of these ions across cell membranes, which is crucial for cellular function, is maintained by the sodium-potassium pump. Sodium concentrations varied from 58.2 mg/L (S6) to 864 mg/L (S3). Although the BIS (IS 10500:2012) does not specify a strict permissible limit for sodium in drinking water, the WHO guidelines suggest that concentrations above 200 mg/L may affect the taste and palatability of water. The highest levels were found at S3 (864 mg/L), S5 (774 mg/L) and S4 (624 mg/L), indicating significant sodium contamination, possibly from industrial discharge or the dissolution of sodium-containing minerals. The lowest concentrations were recorded at S6, S7, S8, S9 and S10 (58.2 - 87 mg/L). Potassium concentrations varied widely, from 2.7 mg/L (S10) to 273 mg/L (S5). The highest potassium

level was recorded at S5 (273 mg/L). The lowest potassium levels were recorded at S10 (2.7 mg/L), S7 (4 mg/L) and S9 (7 mg/L), which fall within the safe drinking limit. The high sodium concentrations observed at locations S3, S4 and S5 may pose health risks, particularly for individuals with hypertension, and can also adversely affect the taste and acceptability of drinking water, while elevated potassium levels could pose risks to individuals with kidney disorders.

Nitrates (NO_3^-)

Nitrates are essential nutrients for plants. They play a crucial role in stimulating the growth of algae, bacteria and aquatic plants in water bodies. Elevated nitrate concentrations can pose risks to human health and the aquatic ecosystem. At the studied locations, nitrate concentration varied from 2.94 mg/L to 9.7 mg/L. All groundwater samples contain nitrate levels below the BIS recommended limit (45 mg/L). Elevated nitrate concentration in potable water may be converted into nitrites, which can interfere with the blood's oxygen-carrying capacity, leading to methaemoglobinaemia (blue baby syndrome) in infants.

Sulphate (SO_4^{2-})

Sulphate is an inorganic anion found in groundwater, particularly in locations with industrial activities, as it is a by-product of many industrial processes such as petroleum extraction and manufacturing. An elevated level of SO_4^{2-} can pose a risk to groundwater quality and the aquatic ecosystem. According to the BIS (IS 10500:2012), the desirable limit for sulphate in drinking water is 200 mg/L, while the permissible limit in the absence of an alternative source is 400 mg/L. In this study, the sulphate concentrations ranged from 16.3 mg/L to 791 mg/L. Sampling locations S2 (431 mg/L), S3 (791 mg/L), S4 (530 mg/L) and S5 (456 mg/L) exhibited sulphate concentrations exceeding the permissible limit of 400 mg/L, suggesting the potential influence of industrial effluents or

geochemical processes in the aquifer system. Concentrations at five study locations were below the desirable limit of 200 mg/L, while one location fell within the permissible limit. Four locations (S2, S3, S4 and S5) exceeded the permissible limit for sulphate (400 mg/L). Elevated sulphate concentrations at these locations may cause laxative effects and gastrointestinal irritation, particularly when consumed over prolonged periods.

Chloride (Cl^-)

Chloride (Cl^-) is an essential electrolyte that plays a key role in maintaining the balance of acids and fluids in the body, nerve and muscle function, and the regulation of blood volume and blood pressure. However, the consumption of water with high concentrations of chloride can cause gastrointestinal problems. The chloride concentrations at four sample locations exceeded the desirable limit (250 mg/L) but they were within the permissible limit (1000 mg/L). Analysis reveals that the Cl^- concentrations at all locations fell within the permissible limit.

Fluoride (F^-)

In the study area, the fluoride concentration varied from 0.18 to 0.92 mg/L. At all ten locations, the fluoride levels were within the desirable limit (i.e., less than 1.0 mg/L), and the water was considered potable. A fluoride content exceeding 1.5 mg/L leads to dental fluorosis, bone weakness, and other health problems.

DO (dissolved oxygen) level

Dissolved oxygen (DO) is an important parameter for assessing water quality, particularly in surface water systems. In groundwater environments, DO concentrations are generally lower due to limited atmospheric interaction and reduced biological activity within aquifers. In this study, the DO values ranged from 3.9 mg/L to 8.2 mg/L, indicating variable oxygen conditions in the groundwater

of the study area. Relatively higher DO values recorded at S4, S7, S8 and S9 may be attributed to recharge from surface water, enhanced groundwater circulation, or localized hydrogeochemical conditions. Overall, the observed DO levels suggest that the groundwater at the studied locations is not significantly affected by organic pollution.

BOD (Biochemical oxygen demand)

BOD is a measure of the quantity of dissolved oxygen used by microorganisms to break down organic matter in water at a specific temperature over a defined period. Domestic, industrial, and agricultural effluents deplete DO levels in water, which can harm aquatic life. Biochemical oxygen demand (BOD) is an indicator of organic pollution in water bodies and is commonly used for assessing surface water quality. In this study, the BOD values were below 1 mg/L at all sampling locations, indicating negligible organic contamination in the groundwater of the study area. The low BOD values suggest a minimal influence of biodegradable organic matter.

COD (Chemical oxygen demand)

Chemical oxygen demand (COD) represents the amount of oxygen required to chemically oxidise organic and inorganic substances present in water. In this study, the COD values

were below 5 mg/L at all sampling locations, which indicates a low level of oxidisable pollutants in the groundwater. These low COD values further suggest a limited impact of organic contamination at the studied locations. However, the DO, BOD and COD parameters are generally more relevant for evaluating surface water quality and were included in this study mainly to assess potential organic contamination from nearby industrial activities.

Correlation analysis of groundwater quality parameters

Pearson's correlation matrix (Table 6) highlights strong interrelationships among the major ions, reflecting the combined influence of geogenic processes and anthropogenic inputs on groundwater chemistry. TDS shows very strong positive correlations with Cl^- ($r = 0.986$), Na^+ ($r = 0.963$) and SO_4^{2-} ($r = 0.931$), indicating that these ions are the principal contributors to groundwater salinity. The Na^+ - Cl^- association suggests contributions from both halite dissolution and anthropogenic sources, such as industrial effluents and sewage. Evaluation of the Na^+/Cl^- ratio indicates deviations from unity at several locations, implying that simple halite dissolution is not the sole process and that additional mechanisms, particularly cation exchange, are taking place.

Table 6. Pearson correlation matrix of various water quality parameters

	pH	TDS	TH	Ca^{2+}	HCO_3^-	Na^+	K^+	NO_3^-	SO_4^{2-}	Cl^-	F^-
pH	1.000										
TDS	-0.299	1.000									
TH	-0.670*	0.759*	1.000								
Ca^{2+}	-0.645*	0.666*	0.980**	1.000							
HCO_3^-	-0.480	-0.208	0.121	0.095	1.000						
Na^+	-0.229	0.963**	0.697*	0.601	-0.284	1.000					
K^+	0.279	0.412	-0.036	-0.147	-0.385	0.558	1.000				
NO_3^-	-0.383	0.656*	0.698*	0.709*	-0.081	0.729*	0.371	1.000			
SO_4^{2-}	-0.359	0.931**	0.882**	0.822**	-0.123	0.931**	0.299	0.768**	1.000		
Cl^-	-0.290	0.986**	0.724*	0.615	-0.212	0.978**	0.470	0.646*	0.927**	1.000	
F^-	0.282	-0.362	-0.348	-0.402	0.302	-0.470	-0.235	-0.793**	-0.419	-0.345	1.000

*Correlation is significant at the 0.05 level (2- tailed), ** Correlation is significant at the 0.01 level (2- tailed)

The enrichment of Na^+ relative to Ca^{2+} and Mg^{2+} indicates the occurrence of ion exchange processes, in which Ca^{2+} and Mg^{2+} in groundwater are replaced by Na^+ adsorbed on aquifer materials. This process is typical of alluvial aquifers and is consistent with the hydrogeological setting of the study area. The strong correlation between SO_4^{2-} and Ca^{2+} ($r = 0.822$) indicates that gypsum dissolution contributes significantly to sulphate levels. However, the elevated SO_4^{2-} concentrations at industrial locations S2 - S5 also indicate anthropogenic inputs. The excellent correlation between TH and Ca^{2+} ($r = 0.980$) confirms that hardness is primarily controlled by the dissolution of carbonate minerals (calcite/dolomite), with an additional contribution from sulphate salts, indicating the presence of permanent hardness.

The simultaneous enrichment of Na^+ , Cl^- and SO_4^{2-} along with high TDS values at selected sites suggests the combined effect of evaporite dissolution and evaporation-induced concentration. Such processes are characteristic of semi-arid environments, where limited recharge and high evaporation rates enhance ionic concentrations in the groundwater. The nitrate levels exhibit moderate to strong correlations with the major ions, indicating mixed anthropogenic sources such as agricultural runoff and wastewater infiltration. In contrast, the strong negative

correlation between NO_3^- and F^- ($r = -0.793$) reflects distinct geochemical controls, with fluoride primarily governed by geogenic processes under alkaline conditions.

Overall, groundwater evolution in the study area is influenced by a combination of carbonate and evaporite mineral dissolution, ion exchange processes, evaporation-driven concentration, and significant anthropogenic inputs from industrial and urban activities. This integrated hydrogeochemical interpretation provides a mechanistic understanding that extends beyond simple correlation analysis.

Water quality index (WQI) analysis

The weighted arithmetic index method was used to calculate the WQI. The suitability of groundwater in the industrial zones was assessed by comparing the results with the BIS standards. The computed WQI values for the ten groundwater samples ranged from 24.39 to 64.58, indicating variations in groundwater quality across the study area. Detailed statistics of the WQI are shown in Table 7. The results show that the majority of the water samples at the studied locations are of “good” quality, while sample S9 (HSI IDC Barhi) is of “poor” quality.

Table 7. Groundwater quality index (WQI) at the various sampling locations

Physicochemical parameters	Sampling locations										Standard value (S_n)	Unit weight ($W_n = k/S_n$)
	1	2	3	4	5	6	7	8	9	10		
pH	7.61	7.8	7.74	7.67	8.27	7.96	7.78	7.83	8.66	8.24	8.5	0.58087
TDS	1724	1185	4765	2923	3132	952	852	926	788	448	500	0.00987
TH	612	890	1165	850	490	485	455	510	256	204	200	0.02469
Ca^{2+}	101	160	188	142	72	100	71.7	85.3	52.8	52	75	0.06583
Mg^{2+}	89.6	144.1	164	123.2	70.7	58.5	68.8	73.5	30.9	18.4	30	0.16458
NO_3^-	8.8	9.7	9.62	8.2	9.1	7.64	2.94	3.21	3.41	4.91	45	0.10972
SO_4^{2-}	304	431	791	530	456	93	86.2	96.6	148	16.3	200	0.02469
Cl^-	362	213	856	624	660	108	158	166	157	27	250	0.01975
*WQI	24.399	32.138	30.148	27.043	49.88	37.702	30.749	32.733	64.581	48.211		

$$*WQI = \sum(Q_n \cdot W_n) / \sum W_n, \text{ where } \sum W_n = 1$$

Although the majority of samples fell into the “good” category based on WQI classification, certain locations exhibited significantly elevated concentrations of parameters such as TDS, Na^+ and SO_4^{2-} . This apparent discrepancy arises from the inherent weighting ($W_n = k/S_n$) scheme of the WAWQI method, where parameters with lower permissible limits are assigned higher weights, as recommended by BIS (2012) and WHO (2017) guidelines and widely adopted in previous studies [25, 26]. Therefore, parameters such as TDS and Na^+ receive lower weights due to their higher permissible limits, rather than their lower health significance. Therefore, WQI values should be interpreted in conjunction with the analysis of individual parameters to avoid underestimation of localized contamination risks. Future studies may incorporate multivariate indices or health-risk assessment approaches to better capture the impact of individual parameters exceeding permissible limits.

The sensitivity evaluation indicates that Mg^{2+} , Ca^{2+} and total hardness (TH) are the most influential parameters affecting the WQI due to their relatively higher unit weights and significant variability. In contrast, TDS, SO_4^{2-} , Na^+ and Cl^- , although dominant in controlling groundwater chemistry and salinity, exert a lower influence on the WQI due to their lower assigned weights. This demonstrates that the WQI is primarily driven by parameters with stricter permissible limits, rather than those with higher absolute concentrations.

The findings of this study are consistent with previous groundwater quality studies conducted in the Haryana and NCR regions. A recent study in the Sonipat district reported a dominance of major ions such as Na^+ , Mg^{2+} , and Cl^- and significant groundwater salinity, which indicates similar hydrochemical characteristics [27]. Similarly, groundwater investigations in the Rohtak district have shown high total hardness, an alkaline nature and elevated concentrations of major ions, including SO_4^{2-} and Na^+ , with a considerable proportion of samples classified as unsuitable for drinking [28]. Studies from other parts of Haryana, such as Mewat (Nuh), also highlight

the role of geogenic processes (e.g., mineral dissolution and ion exchange), along with anthropogenic activities, in controlling groundwater chemistry [29].

These similarities indicate that the elevated TDS, hardness, and ionic concentrations observed at the studied locations are not isolated but reflect a broader regional pattern associated with intensive industrialisation, urbanisation, and hydrogeochemical processes in the NCR-Haryana region. These findings suggest that, while the overall groundwater quality pattern is regionally consistent, site-specific contamination levels require attention and appropriate management.

CONCLUSION

This study provides a comprehensive assessment of groundwater quality at selected industrial locations along the NH-44 corridor in the Sonipat district of Haryana. In this study, ten sampling sites were selected, and sixteen parameters, including BOD and COD, were monitored. The study indicates that the concentrations of calcium, carbonates, bicarbonates, sodium, nitrate, chloride, and fluoride were generally within the permissible limits in most groundwater samples. The higher values of pH, TDS, and TH in some water samples may be attributed to groundwater contamination arising from heavy industrialisation and high population density in the Sonipat district of the NCR, which lies near Delhi, the capital of India.

Based on the calculated WQI values, the majority of the groundwater samples fell within the “good” water quality category, while one sampling location (S9) showed relatively poorer quality, requiring treatment prior to use. The WQI value for Barhi (64.58) indicates that the groundwater requires treatment before consumption. A thorough analysis indicates that the water quality parameters at locations S7 (Murthal) and S10 (HSI IDC, Barhi) meet the standard criteria, while water samples from S3 (Rai) show largest fluctuations in TDS, TH, Mg^{2+} , K^+ ,

SO₄²⁻, and DO levels when compared with the BIS and WHO standards.

Although the WQI classification indicated that most groundwater samples fell within the “good” category, certain locations exhibited elevated concentrations of parameters such as TDS, Na⁺ and SO₄²⁻. This highlights a limitation of the WQI approach, in which parameters with higher permissible limits contribute less to the overall index. Therefore, the WQI results should be interpreted in conjunction with the analysis of individual parameters for a more comprehensive assessment of groundwater quality.

Most of the groundwater quality parameters were within acceptable limits, but elevated levels of TDS, pH, and total hardness at certain locations may pose potential long-term concerns for drinking water suitability if the water is consumed without treatment. Prolonged consumption of water with elevated concentrations of Na⁺, SO₄²⁻, and TDS at certain locations may pose potential health risks, including hypertension, laxative effects, and reduced water palatability, thereby indicating the need for appropriate treatment before consumption. Therefore, regular monitoring and appropriate treatment measures, such as reverse osmosis and water softening systems, are recommended before domestic use. In addition, strict regulation of industrial effluent discharge, regular groundwater quality monitoring, and the adoption of sustainable groundwater management practices, including artificial recharge, can help mitigate groundwater contamination in the studied industrial zones.

The observed variation in groundwater quality among the sampling locations suggests spatial heterogeneity within the studied locations. Although some locations further from Delhi exhibited relatively better water quality indicators, this observation is based on a descriptive analysis and lacks statistical validation. These observations highlight that, despite an acceptable WQI classification at several locations, elevated concentrations of specific parameters may still pose localised health risks and should not be overlooked.

Future studies incorporating spatial gradients or regression analyses are recommended. Future work may include geospatial modelling and distance-based regression to better understand the influence of urban proximity on groundwater quality.

The findings of this study can guide appropriate actions and effective management strategies to safeguard groundwater quality at the studied locations in the Sonipat district, ensuring the long-term sustainability and suitability of local water resources and protecting human health.

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